

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.tech-AI&DS	Date:	

Code of Conduct

I M.KASIF JALAL (192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: kasif jalal _____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device

"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Semantic Representation in Customer Support Chatbots :

The telecom company uses a chatbot to understand customer requests. The semantic analyzer converts natural-language queries into structured representations such as:

- ACTIVATE(Roaming, Customer)
- DEACTIVATE(CallerTune, Customer)
- QUERY(DataBalance, Customer)
- ACTIVATE(5GService, Customer)

The chatbot logs also show that Q2 was incorrectly interpreted.

1. Analyze the semantic representations and identify the action-object relationships

Introduction

Semantic representation is the process of representing the meaning of a natural-language sentence in a structured form that a computer can understand and process.

A chatbot cannot directly execute an operation from a sentence such as:

"Activate international roaming for my number."

It first needs to identify:

1. **What action is requested?**
2. **What object or service is involved?**
3. **Who is performing/requesting the action?**

The semantic representation answers these questions.

The general representation used in the question is:

ACTION (OBJECT, ENTITY)

For example:

ACTIVATE(Roaming, Customer)

means that the **Customer** wants to **activate** the **Roaming** service.

Query 1: "Activate international roaming for my number."

Semantic representation:

ACTIVATE(Roaming, Customer)

Analysis:

- **Action:** ACTIVATE
- **Object:** Roaming
- **Entity:** Customer

The customer wants the telecom system to enable international roaming.

Therefore:

Customer → wants to → ACTIVATE → Roaming

This is a correct semantic representation.

Query 2: "Deactivate caller tune service."

Semantic representation:

DEACTIVATE(CallerTune, Customer)

Analysis:

- **Action:** DEACTIVATE
- **Object:** CallerTune
- **Entity:** Customer

The customer wants the caller tune service to be disabled.

Therefore:

Customer → **wants to** → **DEACTIVATE** → **CallerTune**

This is also a correct semantic representation.

Query 3: "Check my data balance."

Semantic representation:

QUERY(DataBalance, Customer)

Analysis:

- **Action:** QUERY
- **Object:** DataBalance
- **Entity:** Customer

Here, the customer is not asking the system to change a service. Instead, the customer is asking for information.

Therefore:

Customer → **wants to** → **QUERY** → **DataBalance**

The representation correctly distinguishes an information request from an activation/deactivation operation.

Query 4: "Enable 5G service."

Semantic representation:

ACTIVATE(5GService, Customer)

Analysis:

- **Action:** ACTIVATE
- **Object:** 5GService
- **Entity:** Customer

The words "enable" and "activate" have similar meanings in this telecom context.

Therefore:

Enable 5G → ACTIVATE(5GService)

This is a correct semantic interpretation.

Summary

Query	Action	Object	Entity	Representation
Activate international roaming	ACTIVATE	Roaming	Customer	ACTIVATE(Roaming, Customer)
Deactivate caller tune	DEACTIVATE	CallerTune	Customer	DEACTIVATE(CallerTune, Customer)
Check data balance	QUERY	DataBalance	Customer	QUERY(DataBalance, Customer)
Enable 5G service	ACTIVATE	5GService	Customer	ACTIVATE(5GService, Customer)

The important point is that **the action and object together determine the customer's intention.**

2. Determine which query contains semantic interpretation errors

The chatbot logs contain the following information:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Q1

Actual:

Activate Roaming

Predicted:

Activate Roaming

Therefore:

Correct

Q2

Actual:

Deactivate Caller Tune

Predicted:

Activate Caller Tune

Therefore:

Incorrect

The chatbot correctly identified the object:

CallerTune

But it incorrectly identified the action.

Actual semantic representation:

DEACTIVATE(CallerTune, Customer)

Predicted semantic representation:

ACTIVATE(CallerTune, Customer)

The error is therefore an **action-level semantic interpretation error**.

Q3

Actual:

Query Data Balance

Predicted:

Query Data Balance

Therefore:

Correct

Q4

Actual:

Activate 5G Service

Predicted:

Activate 5G Service

Therefore:

Correct

Overall performance

There are:

- 4 total queries
- 3 correctly interpreted
- 1 incorrectly interpreted

Therefore, based only on these four examples:

Semantic accuracy = $3/4 \times 100 = 75\%$

So, the chatbot demonstrates **75% accuracy on the given sample**.

However, this is only the accuracy for the four provided examples and should not be treated as the overall performance of the complete chatbot.

3. Evaluate how meaning representation affects chatbot decision accuracy

Meaning representation is an important intermediate stage between **natural-language input** and **system action**.

The process can be represented as:

User Query

↓

Natural Language Processing

↓

Intent Identification

↓

Entity/Object Identification

↓

Semantic Representation

↓

Decision

↓

Telecom Service Action

For example:

Correct interpretation

User:

"Deactivate caller tune."

NLP system:

DEACTIVATE(CallerTune, Customer)

System action:

Deactivate Caller Tune

This produces the correct result.

Incorrect interpretation

User:

"Deactivate caller tune."

NLP system incorrectly produces:

ACTIVATE(CallerTune, Customer)

System action:

Activate Caller Tune

This produces exactly the opposite of the customer's request.

Therefore, even though the system has correctly identified the entity **CallerTune**, the incorrect action results in an incorrect final decision.

Why semantic representation is important

1. It identifies user intent

The chatbot must understand whether the user wants to:

- Activate
- Q2
- Query
- Modify
- Cancel

a service.

2. It identifies important entities

Examples include:

- Roaming
- Caller Tune
- Data Balance
- 5G Service

3. It supports automatic decision-making

Once the intent and entity are identified, the chatbot can map the request to the appropriate telecom operation.

4. It reduces ambiguity

Words such as:

- enable
- activate
- disable
- deactivate
- check

must be interpreted according to their context.

5. It improves user experience

Correct semantic understanding means that customers receive the requested service without unnecessary interaction.

4. Recommend improvements to enhance semantic understanding

The following NLP improvements can make the chatbot more accurate.

1. Improve intent classification

The model should be trained using a large number of examples for each intent.

For example:

Activation examples:

- Activate 5G
- Enable 5G
- Turn on 5G

Deactivation examples:

- Deactivate 5G
- Disable 5G
- Turn off 5G

This helps the model learn the difference between opposite actions.

2. Use entity recognition

The chatbot should identify telecom entities using **Named Entity Recognition (NER)** or domain-specific

entity extraction.

Example:

"Activate international roaming."

Entities:

Roaming → Telecom Service

3. Use contextual analysis

The meaning of a word depends on its context.

For example:

"Turn off caller tune."

The phrase "turn off" indicates deactivation.

Therefore, the system should not only identify "caller tune" but also understand the relationship between the action phrase and the service.

4. Handle synonyms

Different customers may express the same intent differently.

For example:

- Activate
- Enable
- Turn on

can represent activation.

Similarly:

- Deactivate
- Disable
- Turn off
- Stop

can represent deactivation.

The semantic analyzer should map these expressions to the correct standardized intent.

5. Use negation detection

The system should understand sentences such as:

"Don't activate roaming."

This should not be interpreted as:

ACTIVATE(Roaming)

Instead, the system should recognize the negation.

6. Use dialogue context

Suppose the conversation is:

User: I want to manage my caller tune.

Bot: Do you want to activate or deactivate it?

User: Turn it off.

The word "it" refers to **caller tune**.

Therefore, the system needs dialogue context to generate:

DEACTIVATE(CallerTune, Customer)

7. Confirmation before critical actions

For important service operations, the chatbot can confirm:

"Do you want to deactivate your caller tune service?"

This provides an additional layer of protection against semantic errors.

Conclusion for Q1

The semantic representation successfully converts customer queries into structured actions and objects. Out of the four examples, Q2 contains a semantic error because **DEACTIVATE** was incorrectly interpreted as **ACTIVATE**. This demonstrates that accurate **intent classification, entity recognition, contextual analysis, synonym handling, and negation detection** are essential for reliable NLP-based customer-support chatbots.

Q2. First-Order Predicate Calculus for Smart Manufacturing :

The manufacturing scenario contains four machines:

- $M1 \rightarrow \text{Active}$
- $M2 \rightarrow \text{Active}$
- $M3 \rightarrow \text{Maintenance}$
- $M4 \rightarrow \text{Active}$

The rules provided are:

$\text{Active}(x) \rightarrow \text{Producing}(x)$

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

$\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

1. Represent the production data using First-Order Predicate Calculus

Introduction

First-Order Predicate Calculus (FOPC) is a formal method used in NLP and artificial intelligence to represent facts, relationships, and rules.

It consists of:

- Constants
- Variables
- Predicates
- Logical operators
- Quantifiers

- Implication

In the given manufacturing problem, machines are represented as constants.

Constants

M1, M2, M3, M4

Predicates

Active(x)

means machine x is active.

Maintenance(x)

means machine x is under maintenance.

Producing(x)

means machine x is producing.

Produces(x,y)

means machine x produces product y.

Available(y)

means product y is currently available.

Given machine facts

For M1:

Active(M1)

For M2:

Active(M2)

For M3:

Maintenance(M3)

For M4:

Active(M4)

Rule 1

Given:

$\text{Active}(x) \rightarrow \text{Producing}(x)$

This means:

For every machine x, if x is active, then x is producing.

In quantified form:

$\forall x [\text{Active}(x) \rightarrow \text{Producing}(x)]$

Applying it to M1:

Active(M1)

Therefore:

Producing(M1)

Applying it to M2:

Active(M2)

Therefore:

Producing(M2)

Applying it to M4:

Active(M4)

Therefore:

Producing(M4)

Rule 2

Given:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

This means:

If machine x produces product y AND machine x is active, then product y is available.

In quantified form:

$\forall x \forall y [(\text{Produces}(x,y) \wedge \text{Active}(x)) \rightarrow \text{Available}(y)]$

This rule requires two facts:

1. Machine x produces product y.
2. Machine x is active.

Only when both are true can the system infer:

Available(y)

Rule 3

Given:

$\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

In quantified form:

$\forall x [\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)]$

For M3:

Maintenance(M3)

Therefore:

$\neg \text{Producing}(M3)$

This means M3 is not producing while under maintenance.

2. Analyze which products are currently available

The important rule is:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

However, the provided data does **not give explicit Produces(x,y) facts**.

For example, we are not told:

$\text{Produces}(M1, \text{Gear})$

or:

Produces(M2,Gear)

or:

Produces(M3,Gear)

Therefore, we cannot logically determine a specific product as available from the supplied information.

This is important in First-Order Logic because we should not assume facts that have not been provided.

Hypothetical example

Suppose the manufacturing database contains:

Produces(M1,Gear)

and:

Active(M1)

Then:

$\text{Produces}(\text{M1}, \text{Gear}) \wedge \text{Active}(\text{M1})$

Therefore:

Available(Gear)

But if:

Produces(M3,Gear)

and:

Maintenance(M3)

then:

$\neg \text{Producing}(\text{M3})$

and the system cannot conclude that Gear is available from M3.

Therefore

The correct answer based strictly on the given data is:

The specific products currently available cannot be determined because the relationship between machines and products ($\text{Produces}(x,y)$) has not been provided.

3. Infer whether Gear production is affected by maintenance activities

The maintenance machine is:

M3

Given:

Maintenance(M3)

and the rule:

$\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

we infer:

$\neg \text{Producing}(\text{M3})$

Therefore, M3 is not producing.

Now consider Gear.

If the system has:

Produces(M3,Gear)

then Gear production is affected because M3 is under maintenance.

The reasoning becomes:

Produces(M3,Gear)

Maintenance(M3)

↓

¬Producing(M3)

↓

Gear production by M3 is stopped/affected.

However, because Produces(M3,Gear) is **not explicitly given**, we cannot conclude that M3 produces Gear.

Therefore, the precise logical conclusion is:

Gear production is affected by maintenance only if Gear is produced by M3. The given data does not explicitly establish this relationship.

This distinction is important because First-Order Predicate Calculus does not allow us to assume an unknown relationship as true.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems

Predicate logic is useful in manufacturing because it allows systems to represent real-world situations as logical facts and rules.

Advantages

1. Formal knowledge representation

Machine states can be represented clearly.

Example:

Active(M1)

2. Automated inference

The system can derive new facts.

Example:

Active(M1)

•

Active(x) → Producing(x)

↓

Producing(M1)

3. Rule-based decision making

Industrial systems can use logical rules to determine:

- Machine status

- Production status
- Product availability
- Maintenance effects

4. Explainability

The reasoning can be traced.

For example:

Active(M1)

↓

Producing(M1)

This makes the decision understandable.

5. Consistency

Formal rules reduce ambiguity in decision-making

Limitations

1. Incomplete data

If Produces(x,y) information is missing, product availability cannot be determined.

2. Complex real-world situations

Real manufacturing systems may contain uncertainty, failures, sensor noise, and probabilistic conditions.

3. Rule maintenance

A large industrial system may require thousands of rules.

4. Limited handling of uncertainty

Traditional predicate logic generally represents facts as true or false.

Conclusion for Q2

First-Order Predicate Calculus provides a systematic way to represent manufacturing knowledge and derive conclusions. From the given data, M1, M2, and M4 are producing, while M3 is not producing because it is under maintenance. However, specific product availability and the effect on Gear cannot be definitively determined without Produces(x,y) information.

Q3. Word Sense Disambiguation in E-Commerce Search Engines :

The e-commerce system contains ambiguous words:

- Apple
- Mouse
- Java
- Python

Each word can have multiple meanings. The search context and clicked results help determine the intended meaning.

1. Analyze the contextual information and determine the correct word sense

Introduction

Word Sense Disambiguation (WSD) is an NLP task that determines the correct meaning of an ambiguous word based on its context.

For example:

Apple

could mean:

- Fruit
- Technology brand

The system must use contextual information to identify which meaning the user intended.

Query 1: "Apple accessories"

Possible senses:

1. Fruit
2. Technology/Brand

Clicked result:

iPhone Charger

An iPhone charger is a technology product associated with Apple.

Therefore:

Correct sense = Apple Technology/Brand

Semantic interpretation:

Apple → Technology Brand

Query 2: "Mouse wireless"

Possible senses:

1. Animal
2. Computer device

Clicked result:

Bluetooth Mouse

The words "wireless" and "Bluetooth" strongly indicate a computer peripheral.

Therefore:

Correct sense = Computer Device

Query 3: "Java tutorial"

Possible senses:

1. Island
2. Programming Language

Clicked result:

Coding Lessons

"Coding lessons" are strongly related to programming.

Therefore:

Correct sense = Java Programming Language

Query 4: "Python course"

Possible senses:

1. Snake
2. Programming Language

Clicked result:

Software Development Training

Software development requires programming languages.

Therefore:

Correct sense = Python Programming Language

Final table

Query	Ambiguous Word	Possible Senses	Context	Correct Sense
Apple accessories	Apple	Fruit / Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Mouse	Animal / Computer Device	Bluetooth Mouse	Computer Device
Java tutorial	Java	Island / Programming Language	Coding Lessons	Programming Language
Python course	Python	Snake / Programming Language	Software Development Training	Programming Language

2. Identify the semantic cues used for disambiguation

The system uses **contextual semantic cues**.

Apple

"Accessories" alone can be somewhat ambiguous, but the clicked result **iPhone Charger** provides strong evidence for the technology sense.

Mouse

The word **wireless** is a strong technical cue.

The clicked result **Bluetooth Mouse** confirms the computer-device sense.

Java

The word **tutorial** can occur in many domains, but the clicked result **Coding Lessons** strongly indicates programming.

Python

The word **course** is ambiguous by itself, but **Software Development Training** strongly identifies Python as a programming language.

Types of cues

Lexical cues

Words surrounding the ambiguous word.

Example:

Java + tutorial

Domain cues

Words related to a particular field.

Example:

Python + software development

→ programming domain.

Search-result cues

The clicked result gives additional evidence.

Example:

Apple accessories → iPhone Charger

User behavior cues

In a real system, previous searches and clicked products can also help.

3. Evaluate the impact of incorrect sense selection

Incorrect WSD can seriously affect search performance.

Example: Python

User searches:

"Python course"

Correct meaning:

Programming Language

Suppose the system selects:

Snake

The search engine may return:

- Snake biology
- Reptile information
- Wildlife courses
- Snake photographs

instead of:

- Python programming courses
- Programming tutorials
- Software development courses

Therefore, incorrect WSD results in **low search relevance**.

Effects of incorrect WSD

1. Poor search relevance

Users receive irrelevant results.

2. Lower user satisfaction

Users may not find what they need.

3. Reduced click-through rate

Users are less likely to click irrelevant results.

4. Incorrect recommendations

The system may recommend unrelated products or content.

5. Reduced conversion

In e-commerce, irrelevant recommendations can reduce purchases.

6. Increased search refinement

Users may need to repeatedly modify their queries.

Therefore, WSD directly affects search quality and recommendation performance.

4. Suggest an industrial-scale WSD strategy

A large e-commerce company can use a multi-stage WSD architecture.

Stage 1: Query preprocessing

Perform:

- Tokenization
- Normalization
- Stop-word processing where appropriate

Example:

Python course

→ Python, course

Stage 2: Context extraction

Analyze words surrounding the ambiguous term.

For:

Python course software development

the context strongly indicates programming.

Stage 3: Candidate sense generation

Generate possible meanings.

For Python:

- Snake
- Programming Language

Stage 4: Contextual semantic representation

Represent the query and candidate meanings using contextual language representations.

The system determines which meaning has the strongest relationship with the surrounding context.

Stage 5: Search-result feedback

Use:

- Click data
- Search history
- Product views
- Purchases
- User interactions

to improve sense selection.

Stage 6: Domain knowledge

Use e-commerce knowledge graphs and product categories.

For example:

Python → Programming → Software → Courses

This helps identify the correct sense.

Stage 7: Continuous learning

The model can learn from user feedback.

If users repeatedly click programming courses for "Python course", the system gains stronger evidence that the intended sense is the programming language.

Conclusion for Q3

The given search logs show that context is essential for resolving word ambiguity. The correct senses are **Apple = Technology Brand, Mouse = Computer Device, Java = Programming Language, and Python = Programming Language**. Industrial WSD systems should combine contextual NLP, semantic representations, search behavior, domain knowledge, and continuous learning to improve search and

recommendation accuracy.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems :

The healthcare NLP system processes four sentences:

1. **Doctor prescribed medicine to patient.**
2. **Patient reported severe headache.**
3. **Nurse monitored patient continuously.**
4. **Medicine reduced blood pressure.**

The system assigns semantic roles such as Agent, Instrument, Recipient, and Symptom.

1. Analyze how syntactic structures contribute to semantic role assignment

Introduction

Syntax-driven semantic analysis uses the grammatical structure of a sentence to determine the meaning and semantic relationships between entities.

A syntactic parser identifies:

- Subject
- Verb
- Object
- Complement
- Modifier

These structures help determine semantic roles such as:

- Agent
- Patient
- Recipient
- Theme
- Experiencer
- Instrument
- Symptom

Sentence 1

"Doctor prescribed medicine to patient."

Basic structure:

Subject + Verb + Object + Recipient

Doctor → prescribed → medicine → patient

Semantic roles:

- Doctor → Agent
- Medicine → Theme/Object
- Patient → Recipient

Explanation

The **Doctor** performs the action of prescribing.

Therefore:

Doctor = Agent

The **medicine** is the entity being prescribed.

Therefore:

Medicine = Theme

The **patient** receives the medicine.

Therefore:

Patient = Recipient

Sentence 2

"Patient reported severe headache."

Structure:

Subject + Verb + Object

Patient → reported → severe headache

Semantic roles:

- Patient → Experiencer
- Headache → Symptom

The patient experiences or reports the symptom.

Therefore:

Patient = Experiencer

Headache = Symptom

Sentence 3

"Nurse monitored patient continuously."

Structure:

Subject + Verb + Object + Adverbial Modifier

Nurse → monitored → patient → continuously

Semantic roles:

- Nurse → Agent
- Patient → Theme/Patient
- Continuously → Manner/temporal modifier

The nurse performs the monitoring action.

Therefore:

Nurse = Agent

The patient is the entity being monitored.

Therefore:

Patient = Theme/Patient

Sentence 4

"Medicine reduced blood pressure."

Structure:

Subject + Verb + Object

Medicine → reduced → blood pressure

Semantic roles:

- Medicine → Cause/Agent
- Blood pressure → Affected entity/Theme

The medicine causes a reduction in blood pressure.

Therefore:

Medicine = Cause

Blood Pressure = Affected Entity

2. Determine whether the assigned semantic roles are appropriate

The assessment provides these roles:

Entity Given Role

Doctor Agent

Medicine Instrument

Patient Recipient

Headache Symptom

Now we evaluate each.

Doctor → Agent

This is appropriate.

The doctor performs the action of prescribing.

Doctor → Agent

Patient → Recipient

This is appropriate for:

"Doctor prescribed medicine to patient."

The patient receives the medicine.

Patient → Recipient

Headache → Symptom

This is appropriate.

A headache is a symptom reported by the patient.

Headache → Symptom

Medicine → Instrument

This assignment is **context-dependent**.

In:

"Doctor prescribed medicine to patient."

medicine is more naturally treated as the **Theme/Object** of the prescribing event.

In:

"Medicine reduced blood pressure."

medicine functions as the **Cause** of the reduction.

Therefore, calling medicine an **Instrument** in all cases is not completely accurate.

Improved semantic-role assignment

Sentence	Entity	More appropriate role
Doctor prescribed medicine	Doctor	Agent
Doctor prescribed medicine	Medicine	Theme/Object
Doctor prescribed medicine to patient	Patient	Recipient
Patient reported severe headache	Patient	Experiencer
Patient reported severe headache	Headache	Symptom
Nurse monitored patient	Nurse	Agent
Nurse monitored patient	Patient	Theme/Patient
Medicine reduced blood pressure	Medicine	Cause
Medicine reduced blood pressure	Blood pressure	Affected Entity

Therefore, the original roles are **mostly appropriate**, but **Medicine** → **Instrument** should be interpreted carefully according to the sentence context.

3. Evaluate potential errors that could arise from incorrect parsing

Incorrect syntactic parsing can create serious semantic errors in healthcare applications.

Error 1: Subject identification error

Consider:

Doctor prescribed medicine to patient.

If the parser incorrectly identifies **patient** as the subject, the system may infer:

Patient = Agent

This would completely change the meaning.

Error 2: Object identification error

If the parser identifies the wrong object, the system may not know which medicine was prescribed.

This can cause incorrect extraction of treatment information.

Error 3: Incorrect semantic role assignment

If the system always labels medicine as an Instrument, it may fail to distinguish between:

- Medicine as the object being prescribed

- Medicine as the cause of an effect

This can reduce semantic accuracy.

Error 4: Negation-related errors

Consider a sentence such as:

"Patient did not report headache."

If the NLP system ignores the word "**not**", it could incorrectly extract:

Patient → reported → headache

instead of:

Patient → did NOT report → headache

This is particularly important in healthcare.

Error 5: Temporal interpretation errors

Consider:

"Patient previously had severe headache."

The word **previously** indicates that the symptom occurred in the past.

Incorrect parsing could make the system interpret the headache as a current symptom.

Error 6: Clinical information extraction errors

Incorrect parsing can affect extraction of:

- Medicines
- Diseases
- Symptoms
- Treatments
- Procedures
- Patient conditions
- Medical outcomes

Therefore, syntax-driven semantic analysis must be highly accurate.

4. Recommend methods to improve syntax-driven semantic analysis

1. Dependency parsing

Dependency parsing identifies relationships between words.

For:

Doctor prescribed medicine to patient.

A simplified dependency relationship is:

Doctor → prescribed

medicine → prescribed

patient → prescribed

This helps the NLP system understand who performed the action, what was involved, and who received it.

2. Constituency parsing

Constituency parsing identifies phrases.

For example:

[Doctor] [prescribed [medicine] [to patient]]

This helps identify the grammatical structure of the sentence.

3. Semantic Role Labeling

Semantic Role Labeling (SRL) assigns semantic roles to entities.

For example:

Doctor → Agent

Medicine → Theme

Patient → Recipient

This directly connects syntax with meaning.

4. Medical Named Entity Recognition

Healthcare systems should identify domain-specific entities such as:

- Diseases
- Symptoms
- Medicines
- Body parts
- Medical procedures
- Tests

For example:

"Patient reported severe headache."

NER can identify:

headache → Symptom

5. Context-aware NLP models

A word may have different roles depending on its context.

For example:

"Medicine was prescribed to the patient."

and:

"Medicine reduced blood pressure."

The word **medicine** appears in both sentences but has different semantic relationships.

Context-aware NLP models can help distinguish these differences.

6. Domain-specific training

A general NLP model may not understand all medical terminology correctly.

Training or adapting the system using healthcare datasets can improve:

- Parsing

- Entity recognition
- Semantic-role labeling
- Medical relation extraction

7. Negation detection

Healthcare NLP systems should specifically detect terms such as:

- No
- Not
- Without
- Denies
- Never

Example:

"Patient has no headache."

The system must not extract:

Headache = Present

Instead:

Headache = Absent

8. Temporal analysis

The system should distinguish:

- Past conditions
- Current conditions
- Future conditions

Example:

"Patient had fever last week."

This should be classified as a **past condition**, not a current condition.

9. Human validation

For high-risk medical information, extracted information should be reviewed or validated when necessary.

Overall Conclusion for Q4

Syntax-driven semantic analysis plays an important role in healthcare NLP because syntactic structure provides the foundation for identifying semantic roles. The roles **Doctor** → **Agent**, **Patient** → **Recipient**, and **Headache** → **Symptom** are appropriate, while **Medicine** → **Instrument** is context-dependent and can more accurately be represented as Theme/Object or Cause depending on the sentence.

Accurate **dependency parsing, constituency parsing, semantic role labeling, medical NER, context-aware models, negation detection, and temporal analysis** can improve healthcare NLP systems.

Final Overall Conclusion

The four questions demonstrate four important areas of NLP:

Q1 – Semantic Representation

Converts natural-language user requests into structured meaning representations.

Example:

ACTIVATE(Roaming, Customer)

Q2 – First-Order Predicate Calculus

Represents facts and rules and uses logical inference.

Example:

Active(M1) → Producing(M1)

Q3 – Word Sense Disambiguation

Uses context to identify the correct meaning of ambiguous words.

Example:

Python course → Programming Language

Q4 – Syntax-Driven Semantic Analysis

Uses syntactic structure to determine semantic roles.

Example:

Doctor → Agent

Patient → Recipient

Headache → Symptom

Together, these concepts show how NLP systems move from **language** → **structure** → **meaning** → **reasoning** → **decision-making**. The assessment specifically expects logical interpretation, semantic analysis, predicate representations, and evidence-based conclusions rather than definitions alone.